GLM - Sums of Squares

Lecture 9 - LAS

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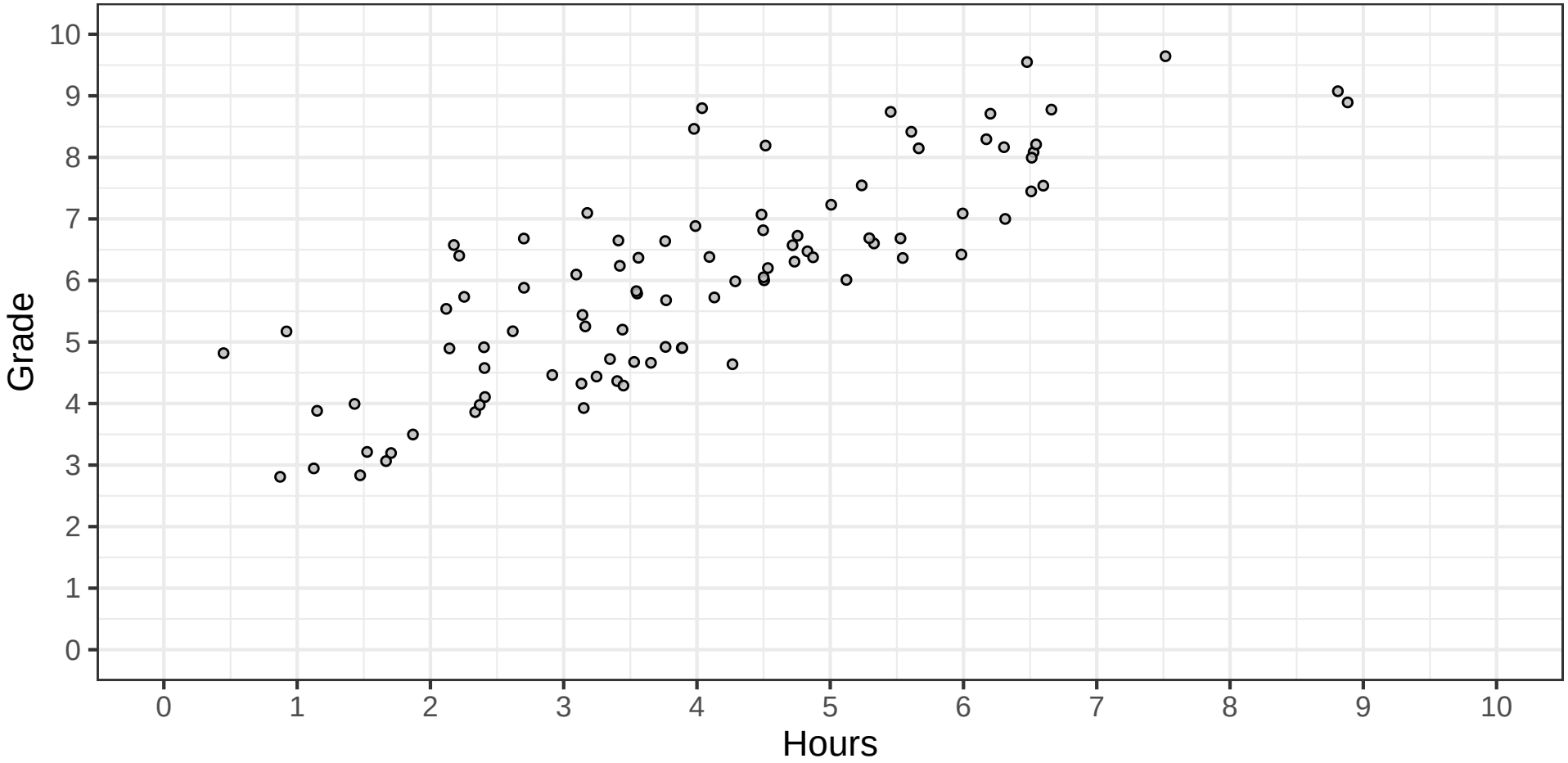
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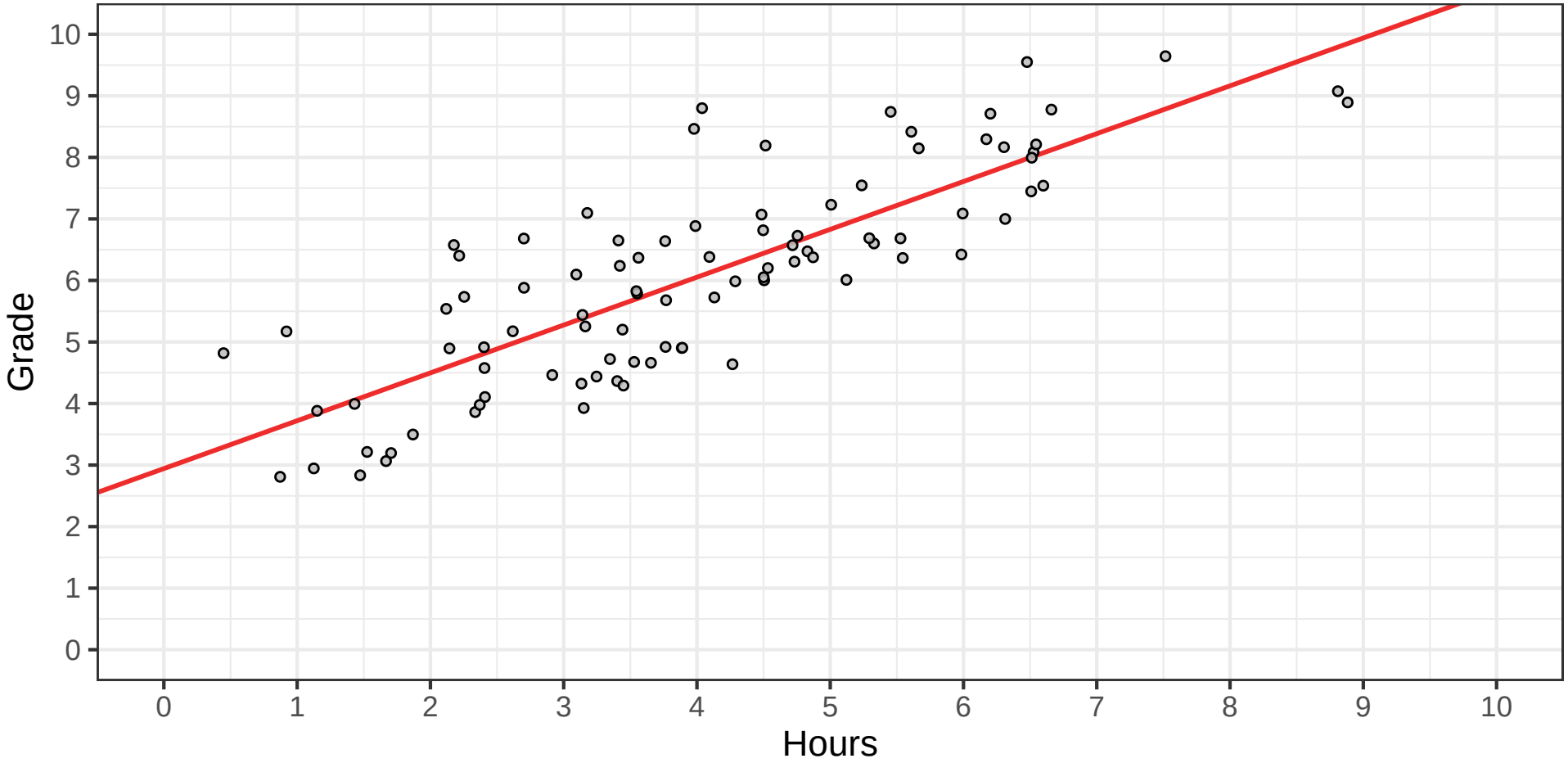
Today

Recap Linear Regression
Sums of Squares
Explained Variance
Correlation
Standardized Regression

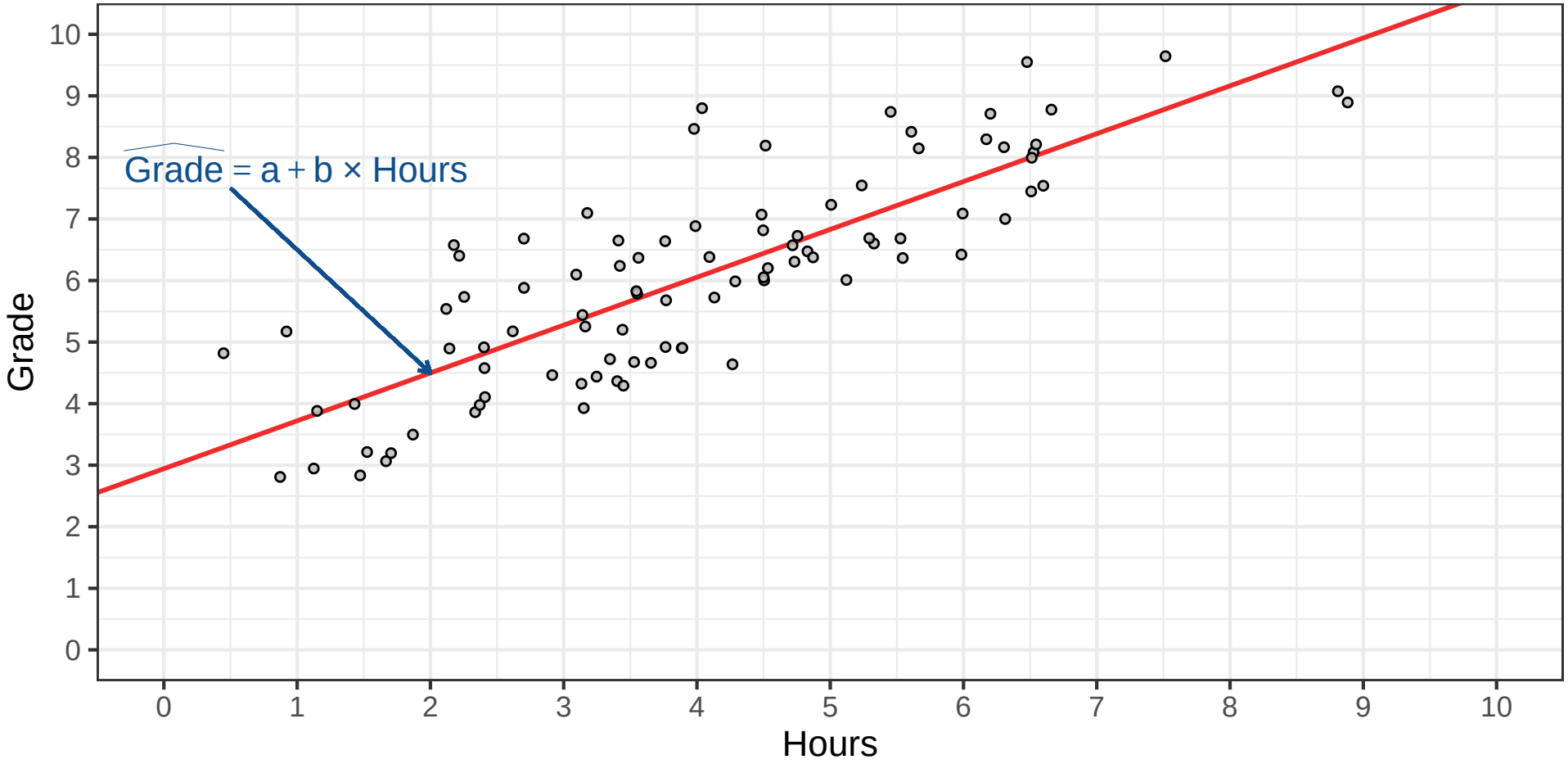
Grades vs. Hours Studied



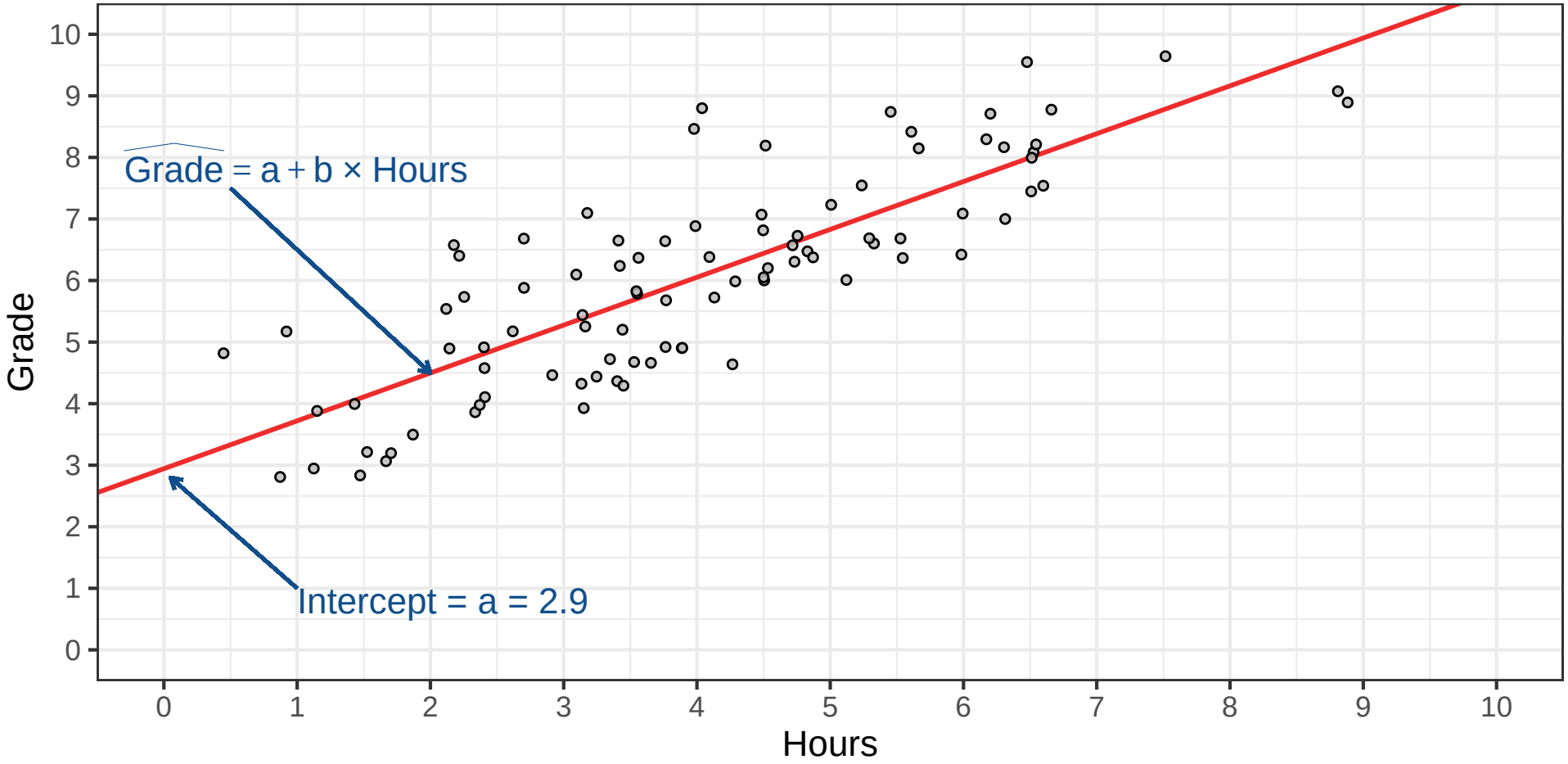
Grades vs. Hours Studied



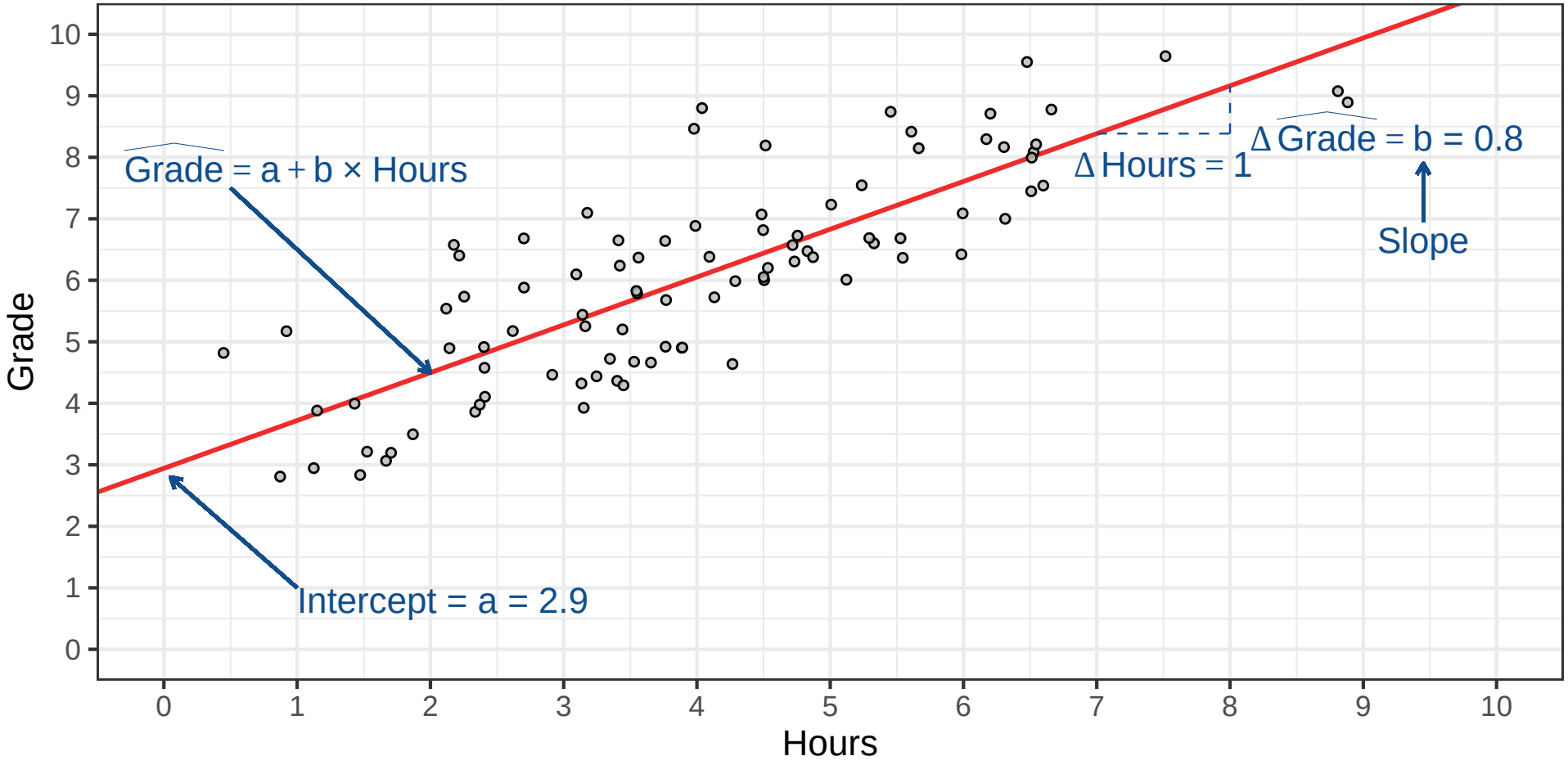
Grades vs. Hours Studied



Grades vs. Hours Studied



Grades vs. Hours Studied



Recap

$$\widehat{\text{Grade}} = \text{Intercept} + \text{Slope} \times \text{Hours studied}$$

In general,

$$\underbrace{y}_{\substack{\text{outcome variable} \\ \text{dependent variable}}} = \underbrace{a}_{\text{intercept}} + \underbrace{b}_{\text{slope}} \times \underbrace{x}_{\substack{\text{predictor variable} \\ \text{independent variable}}} + \underbrace{e}_{\text{residual}}$$

- a, the intercept, is the predicted value when $x = 0$.
- b, the slope, describes how steep the line increases or decreases.
  - $\hat{y}$ increases by b when x increases by 1.

Recap

$$\hat{y}_i = a + b \times x_i, \qquad e_i = y_i - \hat{y}_i$$

Symbol	Interpretation
y_i	Individual i's score on dependent variable y
$\hat{y}_i$	Prediction for individual i's score on dependent variable y
a	Coefficient, intercept of the regression line
b	Coefficient, slope of the regression line
x_i	Individual i's score on independent variable x
e_i	Observed minus predicted outcome for individual i

Grades vs. Hours Studied



We found $a = 2.9$ and $b = 0.8$.

So our equation becomes:

$$\widehat{\text{Grade}} = 2.9 + 0.8 \times \text{Hours studied}$$

Grades vs. Hours Studied

The Sum of Squared Errors (SSE) for these observations is:

$$\text{SSE} = \sum_{i=1}^{3} e_i^2 = 0.8^2 + 0.3^2 + (-0.6)^2 = 1.09$$

Grade	Hours	$\widehat{\text{Grade}}$	e
6.9	4	6.1	0.8
7.2	5	6.9	0.3
7.1	6	7.7	-0.6

Other regression lines

Intercept

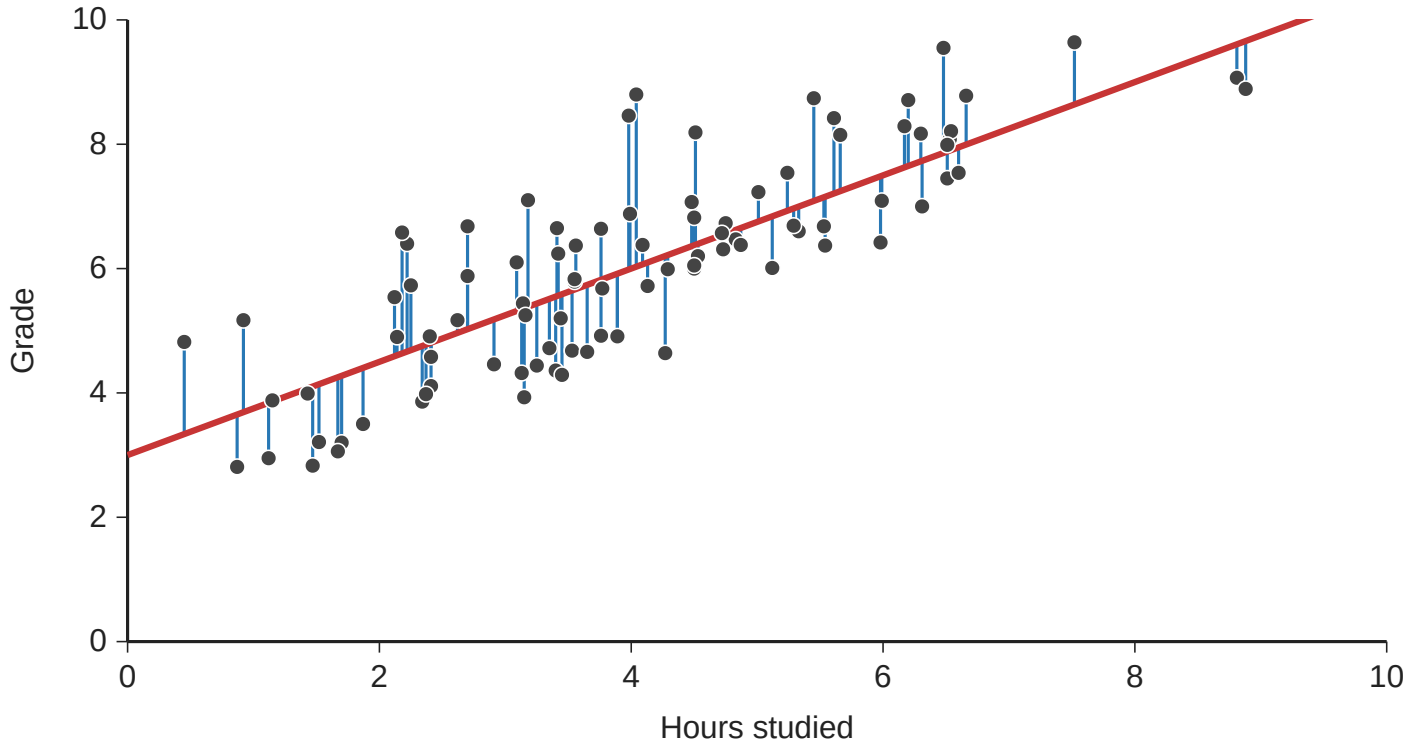
3.0

Slope

0.75

Total squared error:

84.62



Recap



1. The Ordinary Least Squares finds a and b that minimize the SSE.
2. The mean is the best predictor when there's no further relevant information.
3. If we have information about other variables that are associated with the outcome, these can improve the predictions.

Goals for today:

- How much better are the predictions than simply using the mean?
- How does this relate to correlation?



Today

Recap Linear Regression
Sums of Squares
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Prediction error

- OLS finds the straight line with the smallest sum of squared residuals in this sample.
- Literally, this uses “sums of squares”
  - e.g., $e_1^2 + e_2^2 + \cdots + e_N^2$

Ordinary Least Squares

Why squares?

Why not minimize the errors directly?

Last week

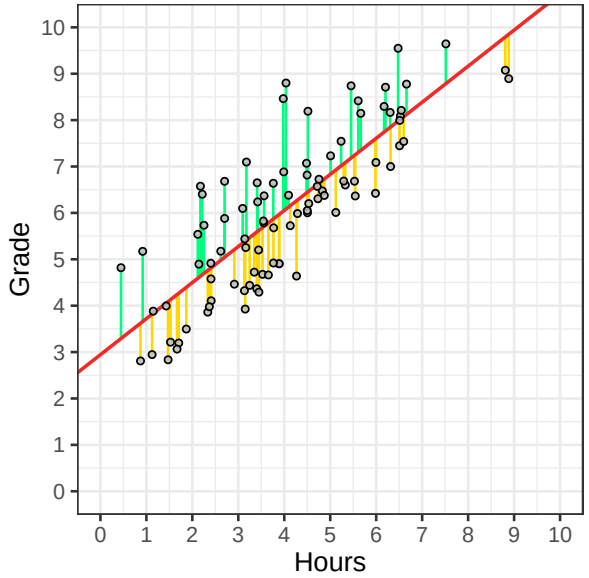
- Suppose: $e_1 = y_1 - \hat{y}_1 = -5$ and $e_2 = y_2 - \hat{y}_2 = 5$
- Then the total error $e_1 + e_2 = 0$.

Ordinary Least Squares

This is not a coincidence!

- Sum of the positive prediction errors: 36.25
- Sum of the negative prediction errors: -36.25
- Sum of all prediction errors: 36.25+-36.25 = 0

The OLS by *design* makes these equal



Solution

Square the errors, so the negatives don’t cancel against the positives.

This means:

- A Sum of Squared Errors (SSE) is always zero or positive.
- A Sum of Squared Errors (SSE) of zero means predictions are perfect.
- Minimizing SSE means finding the regression line that makes the “best” predictions.

Goodness of fit

The OLS gives us the “best” regression line for the outcome given the predictor.

- But how well does it describe the data?
- We can examine the SSE:
  - (+) lower is better
  - (-) the scale is hard to interpret
    - it changes with sample size
    - it depends on the units

We fix the scale by comparing the sum of squared errors to a baseline.

What should the baseline model be?

The null model:

$$y_i = a + e_i$$

- The intercept a, the only parameter is the mean of y.
  - $a = \bar{y}$
  - $\hat{y}_i = \bar{y}$

Comparing to the null model tells us: “How much does this predictor add on top of the intercept?”

Prediction error of the null model

For the null model, this is called the Total Sum of Squares (TSS):

$$\text{TSS} = \sum_{i=1}^{3} e_i^2 = 0.8^2 + 1.0^2 + 1.1^2 = 2.85$$

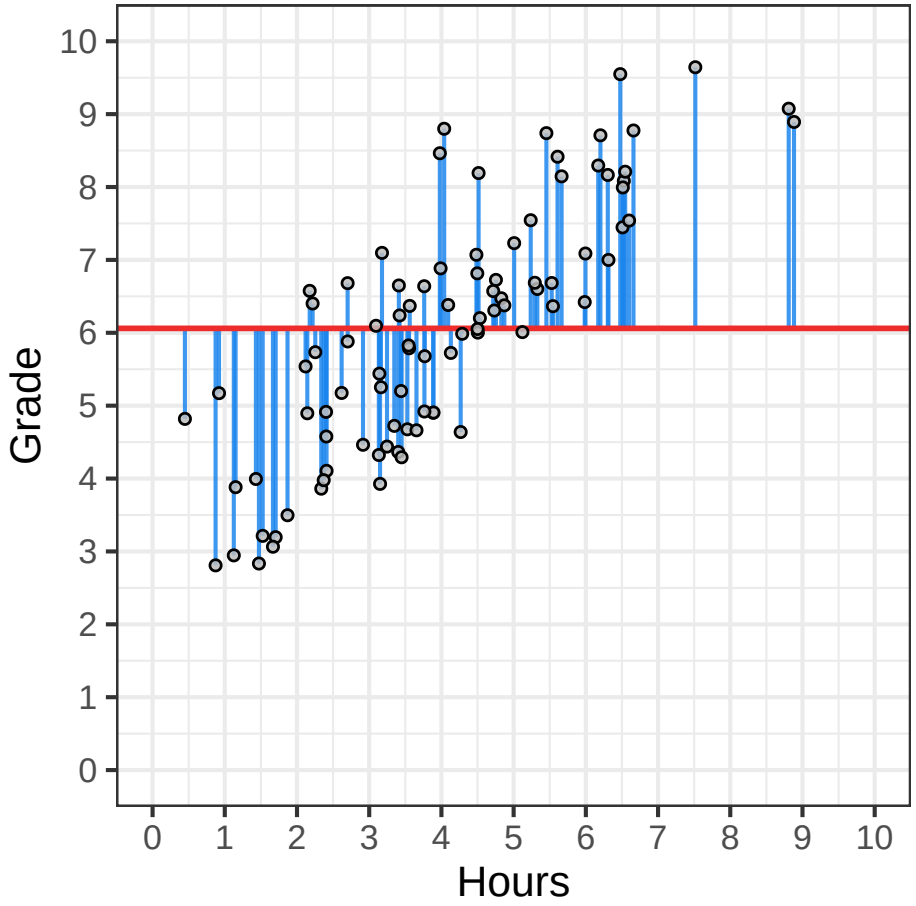
Grade	Hours	$\widehat{\text{Grade}}$	e
6.9	4	6.1	0.8
7.2	5	6.1	1.1
7.1	6	6.1	1.0

Total Sum of Squares

What's the prediction error of the null model for all observations?

$$\begin{aligned}\text{TSS} &= \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^{N} (y_i - \bar{y})^2 \\ &= 255.82\end{aligned}$$

This is the same equation as for the SSE, except we use the predictions from the null model.



How much did our predictions improve?



For all 92 observations:

- Using only the mean: $\mathrm{TSS} = 255.82$.
- Using study hours: $\mathrm{SSE} = 84.18$.

The reduction in squared prediction error is:

$$255.82 - 84.18 = 171.64$$

This is called *Regression Sum of Squares*:

$$\mathrm{RSS} = \mathrm{TSS} - \mathrm{SSE}$$

But is an improvement of 171.64 large?



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What proportion of the error did we remove?



We removed 171.64 from an original squared prediction error of 255.82.

$$\frac{171.64}{255.82} = 0.67$$

Knowing study hours removes 67% of the squared prediction error we had when using only the mean.

We call this proportion R^2:

$$R^2 = \frac{\text{RSS}}{\text{TSS}}$$

A thought experiment

Suppose study hours do not improve our predictions at all.

What would R^2 be?

- No error is removed: $\mathrm{RSS} = 0$.
- Therefore, $R^2 = \frac{\mathrm{RSS}}{\mathrm{TSS}} = \frac{0}{\mathrm{TSS}} = 0$.

Suppose study hours predict every student's grade perfectly.

What would R^2 be?

- All the original error is removed: $\mathrm{RSS} = \mathrm{TSS}$.
- Therefore, $R^2 = \frac{\mathrm{RSS}}{\mathrm{TSS}} = \frac{\mathrm{TSS}}{\mathrm{TSS}} = 1$.

Why call this explained variance?

The original error measures how much grades vary around their mean:

$$\text{TSS} = \sum_{i=1}^{N} (y_i - \bar{y})^2$$

This is also the sum used to calculate the variance:

$$s_y^2 = \frac{\sum_{i=1}^{N} (y_i - \bar{y})^2}{N-1} = \frac{\text{TSS}}{N-1}$$

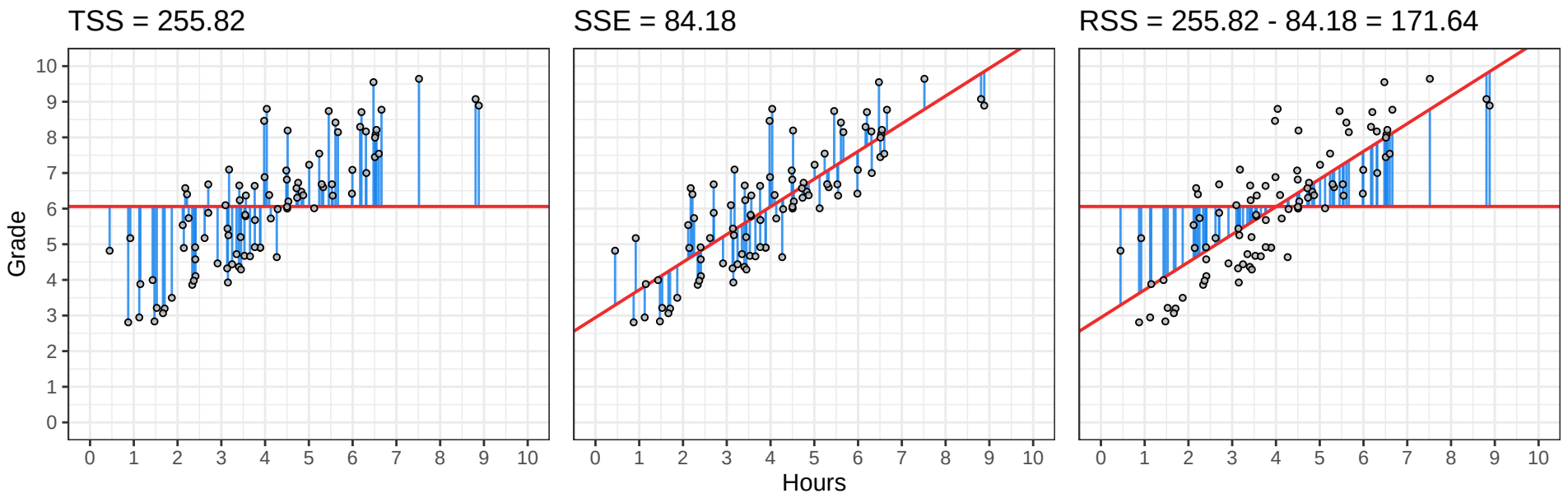
Dividing both RSS and TSS by $N-1$ leaves their ratio unchanged.

So R^2 is also the *proportion of variance explained*.

In our sample, hours studied explains *67% of the variation in grade*.

Total, remaining, and explained variation

- **TSS:** How much variation was there to begin with?
- **SSE:** How much remains unexplained?
- **RSS:** How much has the regression explained?



Three sums of squares: Summary

Quantity	Meaning	Formula
TSS	Total variation	$\sum_{i=1}^{N}(y_i - \bar{y})^2$
SSE	Unexplained variation	$\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$
RSS	Explained variation	$\mathrm{TSS} - \mathrm{SSE}$

The proportion explained is:

$$R^2 = \frac{\text{explained variation}}{\text{total variation}} = \frac{\mathrm{RSS}}{\mathrm{TSS}}$$

Explained Variance: Demo

Mean of y

6.1

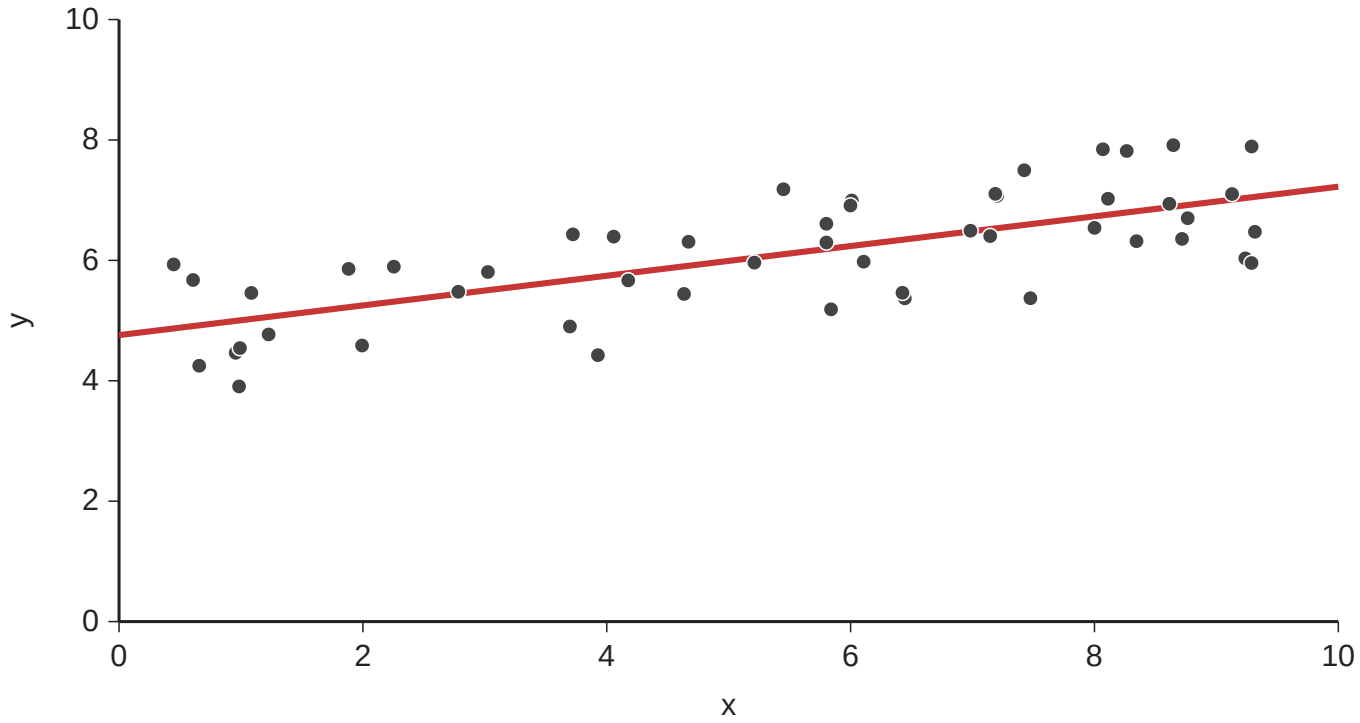
Variance of y

1.0

R^2

0.50

TSS	**49.00**
SSE	**24.50**
RSS	**24.50**



Conclusions:

- Changing the mean of y does not change TSS, SSE, or RSS.
- An explained variance (R^2) of 0 means the slope of the regression is zero $(b = 0)$.
- An explained variance (R^2) above 0 doesn't tell us much about the slope of the regression
  - other than that the slope is not zero.
- An explained variance (R^2) of 1 tells us all observations lie exactly on the regression line.



Today

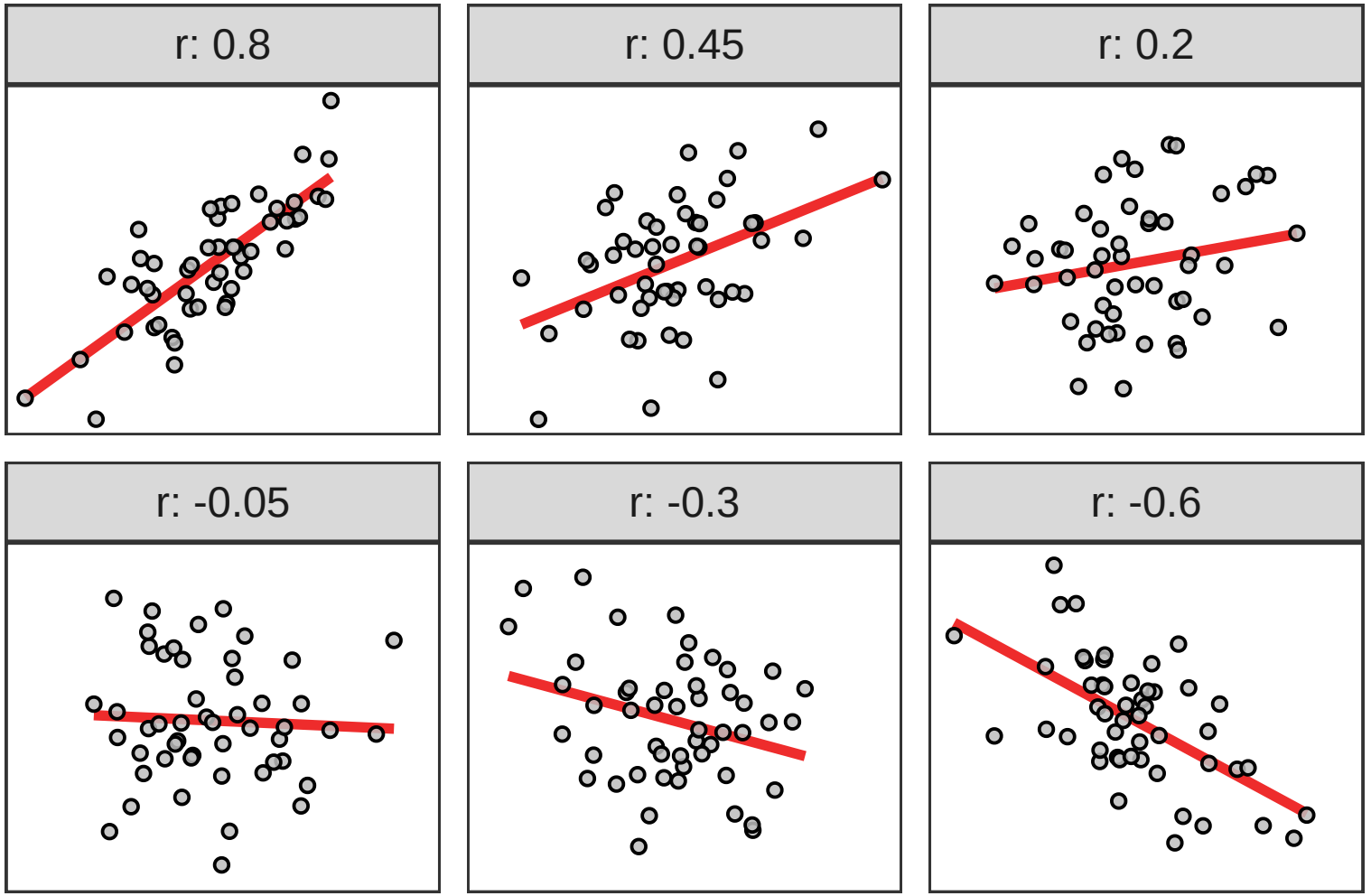
Recap Linear Regression
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Correlation

A standardized measure of the direction and strength of a linear association between two continuous variables

- Standardized: ranges from $[-1,\ 1]$
- Sample correlation: r
- $r = -1$: Perfect negative association
- $r = 0$: No linear association
- $r = 1$: Perfect positive association

Correlation



Correlation



We'll see:

- how to compute a correlation.
- how correlation relates to regression.

Correlation: Computation

In a nutshell:

$$\underbrace{r_{xy}}_{\substack{\text{Correlation between} \\ x \text{ and } y}} = \frac{\overbrace{s_{xy}}^{\text{Covariance between } x \text{ and } y}}{\underbrace{s_x}_{\substack{\text{Standard deviation} \\ \text{of } x}} \times \underbrace{s_y}_{\substack{\text{Standard deviation} \\ \text{of } y}}}$$

Correlation: Computation Example

Using all 92 original observations, $s_x = 1.77$, $s_y = 1.68$, and $s_{xy} = 2.43$.

$$r = \frac{s_{xy}}{s_x \times s_y} = 0.82$$

y	x	$(y - \bar{y})$	$(x - \bar{x})$	$(y - \bar{y})^2$	$(x - \bar{x})^2$	$(y - \bar{y}) \times (x - \bar{x})$
7	6	1	2	1	4	2
4	3	-2	-1	4	1	2
7	3	1	-1	1	1	-1

Correlation: Computation

$$\underbrace{r_{xy}}_{\substack{\text{Correlation between} \\ x \text{ and } y}} = \frac{\overbrace{s_{xy}}^{\text{Covariance between } x \text{ and } y}}{\underbrace{s_x}_{\substack{\text{Standard deviation} \\ \text{of } x}} \times \underbrace{s_y}_{\substack{\text{Standard deviation} \\ \text{of } y}}}$$

The equation for the slope in linear regression is:

$$\underbrace{b}_{\text{Slope}} = \frac{\overbrace{s_{xy}}^{\text{Covariance between } x \text{ and } y}}{\underbrace{s_x}_{\substack{\text{Standard deviation} \\ \text{of } x}} \times \underbrace{s_x}_{\substack{\text{Standard deviation} \\ \text{of } x}}} = \frac{\overbrace{s_{xy}}^{\text{Covariance between } x \text{ and } y}}{\underbrace{s_x^2}_{\text{Variance of } x}}$$

Relation between correlation and regression

$$\underbrace{b}_{\text{Slope}} = \underbrace{r_{xy}}_{\substack{\text{Correlation between} \\ x \text{ and } y}} \times \frac{\overbrace{s_y}^{\substack{\text{Standard deviation} \\ \text{of } y}}}{\underbrace{s_x}_{\substack{\text{Standard deviation} \\ \text{of } x}}}$$

Finding the intercept

The slope sets the steepness. How do we choose the height of the line?

Students studied *4 hours* on average. The average grade was *6.1*.

We found the slope $b = 0.8$. How do we find the intercept?

$$6.1 = a + 0.8 \times 4$$

$$a = 6.1 - 0.8 \times 4 = 2.9$$

In general:

$$a = \bar{y} - b\bar{x}$$

Correlation

Correlation and regression slope both use covariance, but divide it by different quantities.

- Correlation is standardized with respect to x and y.
  - Correlation coefficient has no unit.
  - Correlation of x with y is the same as correlation of y with x.
- Regression coefficient is unstandardized.
  - the parameters have units.
  - it is asymmetric. The regression slope of x on y is different from y on x.
- slope = grade points per hour studied
- intercept = grade points.

Regression is asymmetric



- The regression of Hours studied on Grade does not give the same results as the regression of Grade on Hours studied.
  - unlike the correlation

For $\widehat{\text{Grade}} = a + b \times \text{Hours studied}$, we found $a = 2.9$, $b = 0.8$.

For $\widehat{\text{Hours studied}} = a + b \times \text{Grade}$, we'd find $a = -1.26$, $b = 0.86$.

In both cases, the correlation is $r = 0.82$.

Correlation and explained variance

We've seen that the regression of Grade on Hours had $R^2 = 0.67$.

What does this tell us about the correlation?

$$r^2 = R^2$$

Therefore:

$$r = +\sqrt{R^2}, \qquad \text{if the slope is positive}$$
$$r = -\sqrt{R^2}, \qquad \text{if the slope is negative}$$

Correlation - from Z-scores



There is another way to compute the correlation.

1. Z-transform x and y: $z_x = \frac{x - \bar{x}}{s_x}$, the same for y
2. multiply the values: $z_{xy} = z_x \times z_y$
3. Sum all z_{xy} and divide by $N - 1$.

So we can either:

- Compute the covariance between x and y, standardize by s_x and s_y
- Standardize x and y, then compute the covariance of the z-scores.

For regression we can do the same!



Today

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Standardized Regression Coefficients



Regression slopes depend on the units in which we measure x and y.

Can we express regression coefficients in a way that does not depend on measurement units?

1. Z-transform x and y: $z_x = \frac{x - \bar{x}}{s_x}$, the same for y
2. Regress $\hat{z}_y = a + b \times z_x$

$$\hat{Z}_{\text{Grade}} = a + b \times Z_{\text{Hours studied}}$$

Standardized Regression Coefficients

We obtain $a = 0$ and $b = 0.82$

The intercept is always 0.

The slope equals the correlation we found before!

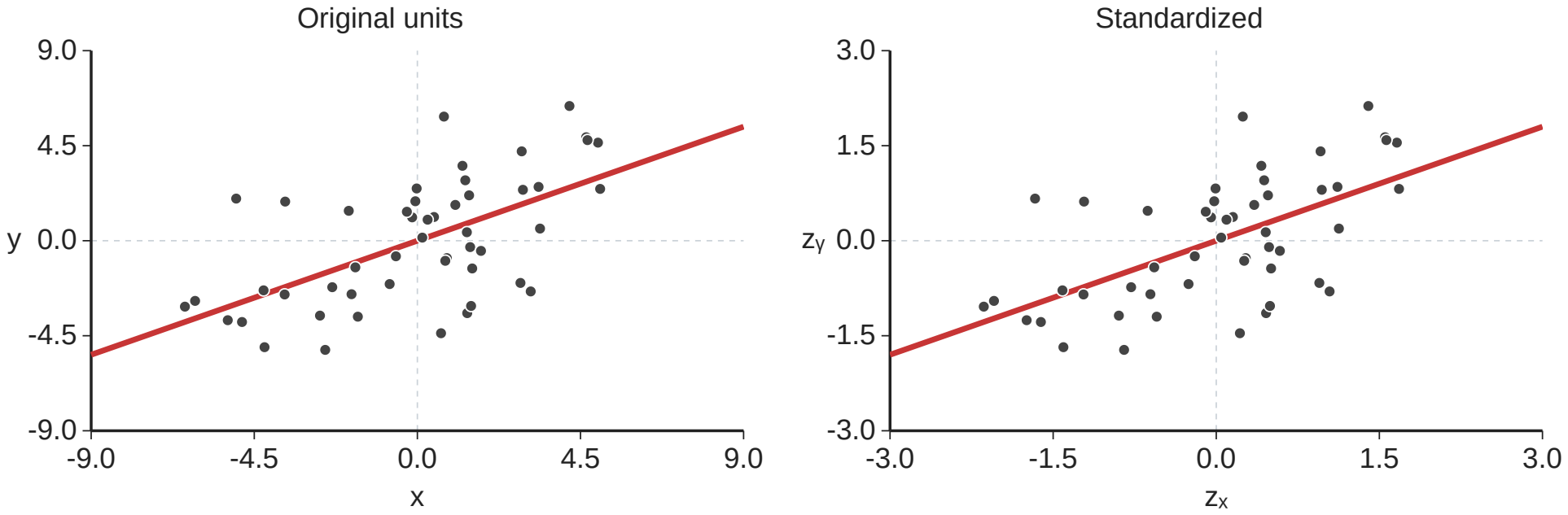
- In bivariate linear regression, the standardized regression slope equals the correlation coefficient.
  - This is no longer true when there is more than one predictor.

Interpretation: A student whose study time is *one standard deviation higher* has a predicted grade *0.82 standard deviations higher*.

Demo: Units and Standardized Regression

$\bar{x}$ 0.0 $\bar{y}$ 0.0 s_x 3.0 s_y 3.0 Correlation r 0.60

Intercept a	Slope b	$R^2 = r^2$	Std. intercept	Std. slope
0.00	**0.60**	**0.36**	**0.00**	**0.60**



Both panels show the same 50 observations

What did the demo show?

- Changing the means or measurement units can change the original regression coefficients.
- These changes leave the correlation and R^2 unchanged.
- After converting both variables to z-scores, the intercept is 0 and the slope equals r.

Summary

- TSS: total variation in y around its mean.
- SSE: variation left unexplained by the regression.
- RSS: variation explained by the regression: $\mathrm{RSS} = \mathrm{TSS} - \mathrm{SSE}$.
- $R^2 = \mathrm{RSS}/\mathrm{TSS}$: the proportion of variance explained.
- r: the direction and strength of the linear association.

For bivariate regression: $R^2 = r^2$, and the standardized slope equals r.

Next Week

Testing and checking a regression model

- Test regression coefficients using t-tests.
- Test the regression model using the F-test.
- Check linearity, normality of residuals, homoscedasticity, and independence.